Challenge Description:

We intercepted two suspicious files from the Chum Bucket's servers. The SpongeBob file looks damaged... but something feels off. Patrick probably knows more than he lets on — he always does. Can you recover the secret formula before Plankton does?

Solution:

Here we see two files, spongebob.jpeg and patrick.bmp

spongebob.jpeg seems to be corrupted, so we open it with https://hexed.it/ to repair it:

```
-Untitled-   spongebob.jpeg
00000000   00 00 FF E1 0F FE 45 78 88 89 00 00 4D 4D 00 2A   .. ß.▪Exêë..MM.*
00000010   00 00 00 08 00 06 01 12 00 03 00 00 00 01 00 01   ................
00000020   00 00 01 1A 00 05 00 00 00 01 00 00 00 56 01 1B   .............V..
```

We can see that jpeg magic numbers are FF D8 FF but are corrupted to 00 00 FF
So we repair it :

```
-Untitled-   spongebob.jpeg
00000000   FF D8 FF E1 0F FE 45 78 88 89 00 00 4D 4D 00 2A   ╪ß.▪Exêë..MM.*
00000010   00 00 00 08 00 06 01 12 00 03 00 00 00 01 00 01   ................
00000020   00 00 01 1A 00 05 00 00 00 01 00 00 00 56 01 1B   .............V..
```

Then we get this image:

We can notice the word -> “krabby-patties” written faintly inside it. This might be a password for something, a common and famous password based encryption tool for images is steghide.

Use that password with steghide on patrick.bmp to get:

```
mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads$ steghide extract -sf patrick.bmp
Enter passphrase:
wrote extracted data to "output.mp3".
mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads$
```

We get output.mp3.
Now it is natural to be confused as to how we can decrypt contents inside this, for this, if we run strings on patrick.bmp, we get:

```
mp3stego_with_plankton@2026
mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads$
```

At the end…
This means that it has been encrypted using mp3stego and the password that was used is plankton@2026. So use mp3stego to decode, with plankton@2026 as the password:

Download mp3stego from here (clone it):
https://codeberg.org/fabienpe/MP3Stego

Then decode:

```
mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads/MP3Stego/MP3Stego$ wine Decode.exe -X -P plankton@2026 output.mp3
MP3StegoEncoder 1.1.19
See README file for copyright info
Input file = 'output.mp3'  output file = 'output.mp3.pcm'
Will attempt to extract hidden information. Output: output.mp3.txt
the bit stream file output.mp3 is a BINARY file
HDR: s=FFF, id=1, l=3, ep=off, br=9, sf=0, pd=1, pr=0, m=0, js=0, c=0, o=0, e=0
alg.=MPEG-1, layer=III, tot bitrate=128, sfrq=44.1
mode=stereo, sblim=32, jsbd=32, ch=2
[Frame 1302]Avg slots/frame = 417.640; b/smp = 2.90; br = 127.902 kbps
Decoding of "output.mp3" is finished
The decoded PCM output file name is "output.mp3.pcm"
mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads/MP3Stego/MP3Stego$ ls
Decode.exe  Encode.exe  hidden_text.txt  output.mp3      output.mp3.txt  tables
Decoder     Encoder     MP3Stego.sln     output.mp3.pcm  svega.wav
mrsuave@mrsuave-HP-Laptop-15-bs0xx:~/Downloads/MP3Stego/MP3Stego$ cat output.mp3.txt
TACHYON{killing_1t_$p0ng3b0b!}
```

Flag:
TACHYON{killing_1t_$p0ng3b0b!}